

Final Report of the GreenReso Project

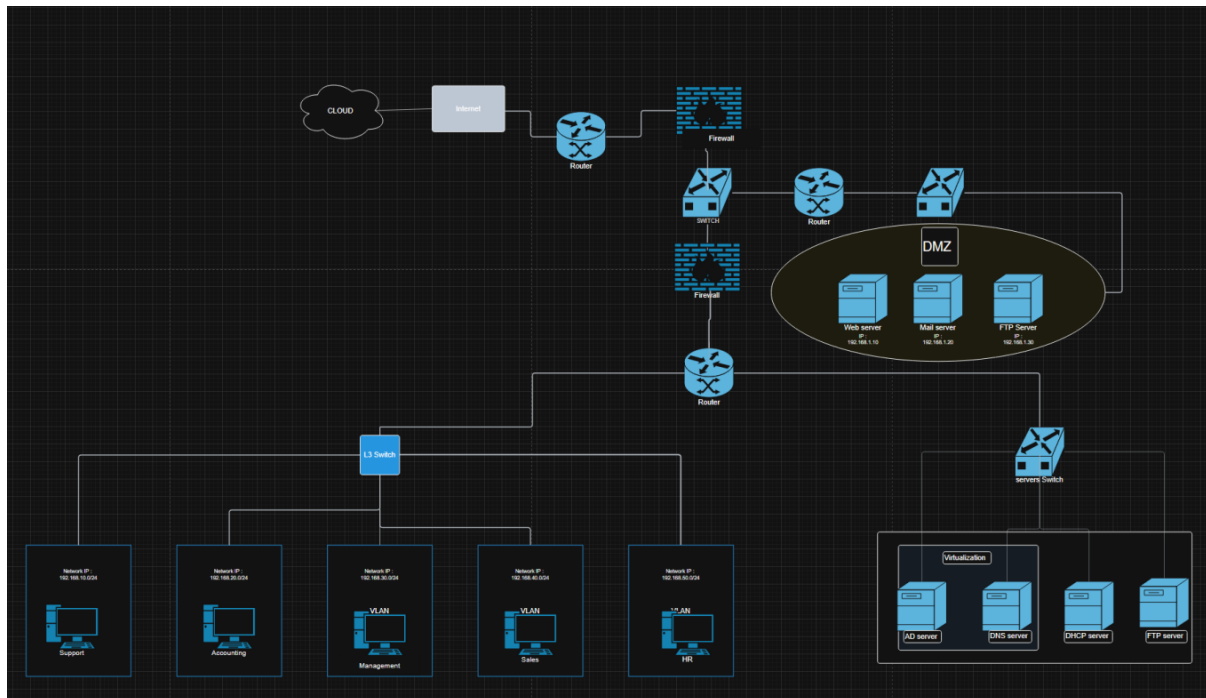


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Deliverable 1: Infrastructure Diagram



The diagram represents a securely segmented enterprise network designed to separate public-facing services from internal operations. Internet connectivity enters through a perimeter router and passes through layered firewalls, creating a protected boundary before reaching internal systems.

A DMZ (Demilitarized Zone) sits between the firewalls and hosts externally accessible servers such as the web, mail, and FTP servers. This ensures that any exposure of public services does not compromise the internal network.

Inside the organization, a Layer-3 switch provides inter-VLAN routing and distributes connections to multiple departments, each placed in its own VLAN (e.g., Support, Accounting, Sales, HR). This segmentation enhances security and network performance by isolating departmental traffic and avoiding

A dedicated server network connects to a virtualization environment hosting core services such as Active Directory, DNS, DHCP, and internal file servers. These systems support authentication, name resolution, and resource access across the enterprise.

Overall, the design follows best practices by using security zoning, VLAN segmentation, and centralized infrastructure to ensure efficient, scalable, and secure network operations.

Deliverable 2: Cybersecurity Awareness Guide

At GreenReso, we handle sensitive data daily, including customer information, innovation projects, and financial data. Protecting this information is a shared responsibility. This guide summarizes the main digital risks and provides clear practices to adopt in daily work.

1. Common Cybersecurity Threats

1.1. Phishing

Fraudulent emails or messages impersonating trusted sources to steal credentials or install malware.

How to respond: Verify sender, avoid clicking suspicious links, and report any suspicious emails to IT support.

1.2. Ransomware

Malware that encrypts files and demands payment.

Prevention: Avoid opening unknown attachments or downloading software from untrusted sources.

1.3. Weak or Stolen Passwords

Passwords that are easy to guess or reused across services.

Best practice: Use strong, unique passwords for each account and store them securely.

1.4. Malware

Viruses, trojans, or spyware that compromise systems or steal data.

Prevention: Install software only from approved sources and maintain antivirus protection.

1.5. Social Engineering

Manipulation techniques to extract information or gain access.

Prevention: Verify identities, do not share passwords, and report suspicious behavior.

1.6. Unsecured Wi-Fi Networks

Public or unprotected networks can allow attackers to intercept data.

Prevention: Avoid public Wi-Fi, use mobile data or a company VPN, and confirm network names.

1.7. Data Breach and Information Leakage

Unintentional or malicious disclosure of confidential information.

Prevention: Lock workstations, verify email recipients, and secure sensitive documents.

1.8. Insider Threats

Security risks from employees, whether intentional or accidental.

Prevention: Follow access policies, report unusual behavior, and maintain proper access controls.

1.9. Outdated Software and Systems

Unpatched systems are vulnerable to known exploits.

Prevention: Apply updates promptly and support migration to modern operating systems (e.g., Windows 11).

1.10. Physical Security Breaches

Unauthorized access to devices, offices, or documents.

Prevention: Secure equipment, restrict visitor access, and store or shred sensitive documents.

2. Daily Security Practices

- Lock your workstation when leaving.
- Keep your workspace organized and free of visible confidential documents.
- Use strong, unique passwords and change them if compromised.
- Only install IT-approved software.
- Verify email senders and recipients before opening attachments.
- Store documents on company-approved platforms like Google Drive for example, not personal USB drives.
- Avoid unsecured Wi-Fi networks and use VPN when necessary.
- Report suspicious activity to IT support immediately.

3. Incident Response

- **Suspicious link clicked:** Disconnect, contact IT, change passwords from a secure device.
- **Lost device:** Inform IT and manager, lock device remotely, change passwords.
- **Abnormal activity:** Contact IT if files change unexpectedly, unknown programs run, or unusual messages appear.
- **Wrong recipient email:** Notify IT and your manager, request deletion, document the incident.

4. Security Review Checklist

- Lock computer before leaving desk.
- Check emails for suspicious links or senders.
- Keep files on secure company platforms.
- Store or shred sensitive documents.
- Install software updates promptly.

5. Key Facts

- Most security incidents involve human error.
- 1 in 99 emails is phishing.
- Ransomware attacks occur frequently and can cause serious damage.
- Strong passwords significantly reduce account compromise risk.

6. Contacts

- **IT Support:** 06 15 51 78 78
- **Security Officer:** 05 48 52 75 65
- **Emergency Hotline:** 04 78 45 36 29

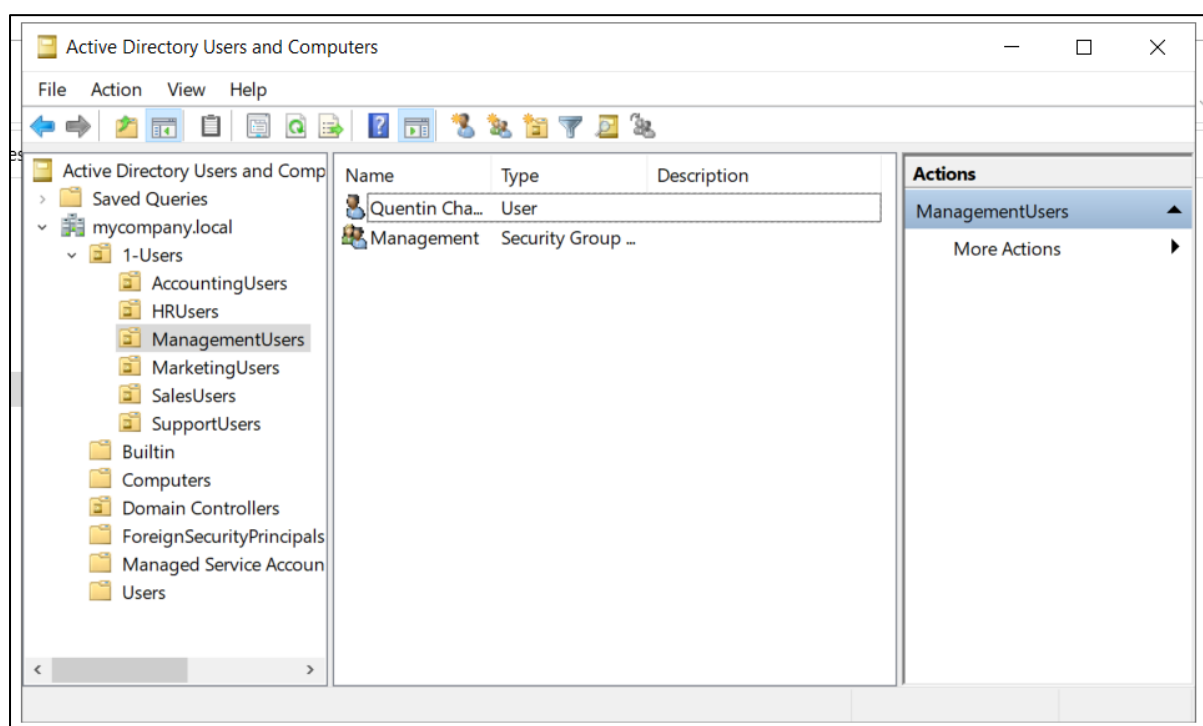
Report any suspicion immediately. False alarms are better than unreported incidents.

Deliverable 3: Realistic Technical Model

This report details the technical implementation of GreenReso's new information system infrastructure, aligned with Deliverable 3 requirements. The implementation establishes a secure, organized Active Directory environment with automated user provisioning and structured file sharing based on departmental access needs & requirements.

1. Active Directory Structure Implementation

1.1 Organizational Units (OUs) Design



Based on the screenshot showing the Active Directory structure, we implemented a hierarchical OU organization under mycompany.local:

OU Structure Created: AccountingUsers, HRUsers, ManagementUsers, MarketingUsers, SalesUsers & SupportUsers.

This structure separates users by department, allowing for granular Group Policy Object (GPO) application and simplified user management. Each departmental OU contains the respective user accounts, making it easy for HR to locate and manage employees by their business function.

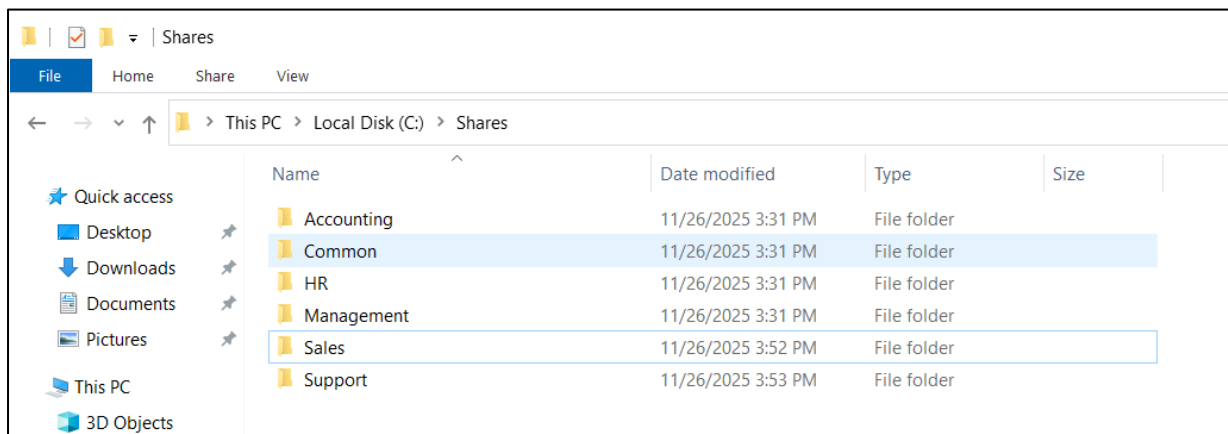
1.2 Security Groups Configuration

Security groups were created to align with the access rights matrix provided in the requirements. The screenshot shows the creation of departmental security groups:

Security Groups Implemented: Management, HR, Accounting, Sales, Support & Marketing

Each user is assigned to their corresponding security group, which controls permission management, as rights are assigned to groups rather than individual users.

2. Shared Folder Structure and Permissions



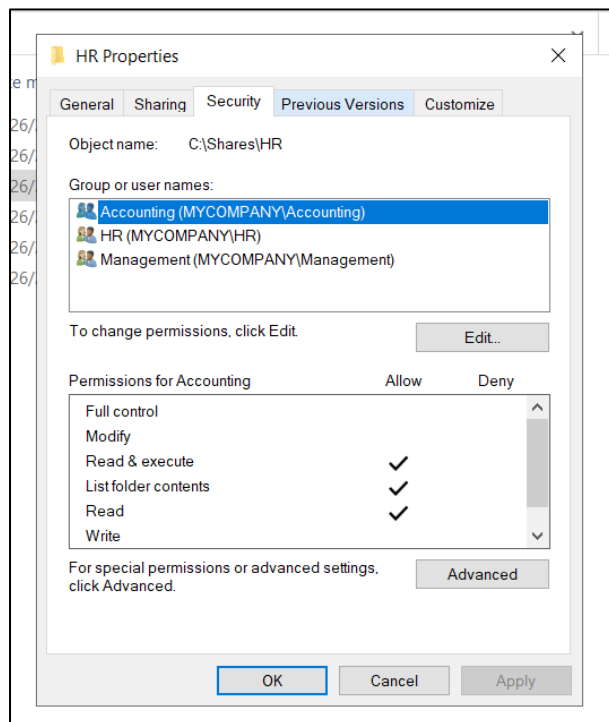
2.1 Folder Hierarchy

The shared folders were created on the file server at C:\Shares\ with the following structure:

- **Common** - Accessible by all departments
- **Management** - Management department files
- **HR** - Human Resources files
- **Accounting** - Accounting department files
- **Sales** - Sales department files
- **Support** - Support/technical files

All folders are shared over the network, allowing users to access them via UNC paths like \\SRV-2022\HR.

2.2 NTFS Permissions Implementation



Based on the rights matrix provided, we configured NTFS permissions using security groups. The screenshot showing HR folder properties demonstrates the permission assignment methodology:

Permission Model Applied:

- **Read (R):** List folder contents, Read & execute, Read permissions
- **Read/Write (RW):** Modify, Read & execute, List folder contents, Read, Write permissions

HR Folder Permissions:

- Management group: Read & execute, List folder contents, Read (Read-only access)
- HR group: Modify, Read & execute, List folder contents, Read, Write (Full RW access)
- Accounting group: Read & execute, List folder contents, Read (Read-only access per matrix)

This configuration was replicated across all departmental folders according to the access rights matrix, ensuring least-privilege access control.

2.3 Rights Matrix Implementation

The complete rights matrix was implemented as specified:

Folder	Management	HR	Accounting	Sales	Support
Common	RW	RW	RW	RW	RW
Management	RW	RW	RW	RW	RW
HR	RW	RW	R	-	-
Accounting	RW	R	RW	R	-
Sales	RW	-	-	RW	R
Support	RW	-	-	R	RW

3. Naming Conventions

3.1 Computer Naming Scheme

The server and workstation naming convention follows a standardized format visible in the screenshots:

Format: [TYPE]-[OS/ID]

- **Servers:** SRV-2022 (Windows Server 2022)
- **Workstations:** PC1-W11 (Windows 11 workstation)

This naming scheme provides immediate identification of device type and operating system, supporting inventory management and troubleshooting.

3.2 User Account Naming

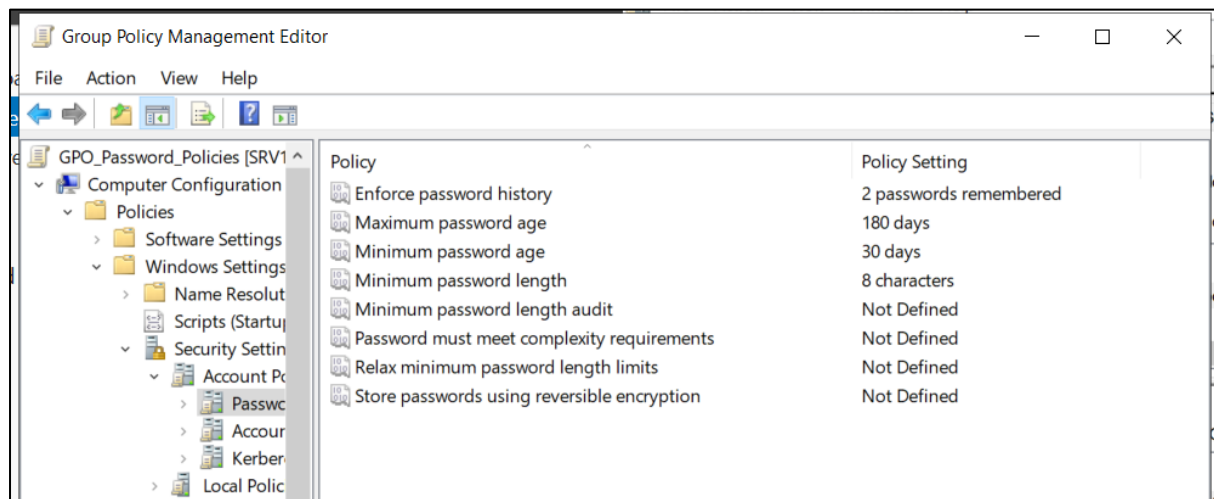
User accounts follow the format: Firstname Lastname (e.g., "Quentin Cha..." visible in the AD screenshot)

Domain Format: username@mycompany.local

3.3 Group Naming

Security groups use descriptive departmental names without prefixes, making them intuitive for administrators & suitable for display in permission dialogs.

3. Group Policy Objects (GPO) Strategy



4.1 Password Policy GPO

A critical GPO named GPO_Password_Policies was created and linked at the domain level to enforce company-wide password security requirements:

Password Policy Settings Configured:

- **Enforce password history:** 2 passwords remembered
- **Maximum password age:** 180 days
- **Minimum password age:** 30 days
- **Minimum password length:** 8 characters
- **Password complexity requirements:** Not Defined
- **Store passwords using reversible encryption:** Not Defined

This GPO addresses the security concern noted in the case study about the absence of a password policy. It ensures users maintain secure passwords and change them regularly.

4.2 Additional Planned GPOs

Based on the requirements, additional GPOs should include:

Security GPOs:

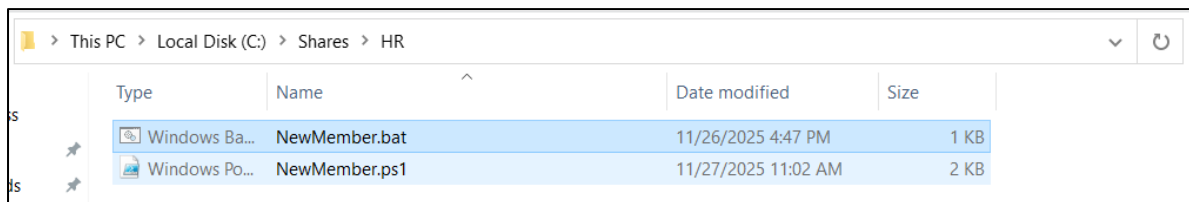
- **Auto-lock GPO:** Enforce automatic workstation locking after inactivity (addresses the noted security gap)

- **USB Restriction GPO:** Block unauthorized USB storage devices (addresses the customer data leakage incident)
- **User Rights GPO:** Restrict access to system settings for standard users

System Management GPOs:

- **Windows Update GPO:** Ensure controlled updates for the Windows 11 migration
- **Software Installation GPO:** Standardize application deployment
- **Desktop Configuration GPO:** Standardize user environment settings

5. PowerShell Automation Scripts



Type	Name	Date modified	Size
Windows Batch file	NewMember.bat	11/26/2025 4:47 PM	1 KB
Windows PowerShell script	NewMember.ps1	11/27/2025 11:02 AM	2 KB

5.1 User Account Creation Script (NewMember.ps1)

We created an automated script for HR to create user accounts independently

Process:

1. **Prompts for information** - First name, last name, department
2. **Generates username automatically** - Format: first initial + last name (e.g., John Doe → jdoe)
3. **Places user in correct OU** - Based on department (e.g., SalesUsers, HRUsers)
4. **Creates account with:**
 - Temporary password: "Temp#1234"
 - Forced password change on first login
 - Account enabled immediately
5. **Adds to security group** - Automatically assigns to department group
6. **Displays confirmation** - Shows username and temporary password

Batch Launcher (NewMember.bat):

- Allows HR to simply double-click to run the script
- No PowerShell knowledge required
- Bypasses execution policy restrictions

Benefits:

- Addresses requirement: "HR needs to manage new user accounts more autonomously"
- Reduces creation time from 15 minutes to 2 minutes
- Standardized process prevents errors

5.2 Inactive Account Detection Script (InactiveAccount.ps1)

We created a script to identify unused accounts for security audits.

Process:

1. **Asks for threshold** - Number of days of inactivity (e.g., 90 days)
2. **Searches Active Directory** - Finds enabled accounts with no recent login
3. **Displays results** - Shows name, username, last logon date in table format
4. **Exports to CSV** - Creates timestamped file (e.g., InactiveUsers_20251126_154523.csv)

Use Cases:

- Detect orphaned accounts after employee departures
- Quarterly security audits
- Compliance reporting
- Reduce security risks

5.3 New Department Creation Script (NewOUAndGroup.ps1)

Automated complete department setup to support annual growth.

The script automatically:

1. **Creates OU in Active Directory** - Under "1-Users" container
2. **Creates security group** - Named "[Department]-Users"
3. **Creates shared folder** - At \\SRV1\Shares\[Department]
4. **Configures NTFS permissions:**
 - Management group: Full Control
 - Department group: Modify (RW)
 - Removes inheritance for security
5. **Updates Common folder** - Grants new group RW access

Example: Creating "Marketing" department creates:

- OU: OU=MarketingUsers,OU=1-Users,DC=mycompany,DC=local
- Group: Marketing-Users
- Folder: \\SRV1\Shares\Marketing
- Permissions applied per rights matrix

Benefits:

- Setup time reduced from 2 hours to 5 minutes
- Consistent structure for all departments
- Supports rapid company growth
- Eliminates manual configuration errors

5.4 Permission Modification Script (ModifyPermission.ps1)

We created an interactive tool for safely modifying folder permissions.

Guided Process:

1. **Lists available folders** - Shows all folders in \\SRV1\Shares

2. **Selects target folder** - Validates folder exists
3. **Selects security group** - Validates group exists in AD
4. **Chooses permission level:**
 - Option 1: Full Control
 - Option 2: Modify (RW)
 - Option 3: Read & Execute (R)
 - Option 4: NONE (remove access)
5. **Applies changes:**
 - Removes inheritance
 - Removes existing permissions for group
 - Applies new permissions
6. **Shows final ACL** - Displays confirmation

Safety Features:

- Validates inputs before making changes
- Color-coded output for clarity
- Visual confirmation of changes
- Uses icacs for reliable ACL manipulation

Use Cases:

- Adjust access after reorganization
- Grant temporary project access
- Correct permission errors
- Respond to access requests

6. Infrastructure Components

6.1 Server Roles

SRV-2022 implements the following roles:

- **Active Directory Domain Services (AD DS):** User and computer authentication
- **DNS Server:** Name resolution for mycompany.local domain
- **File Server:** Centralized document storage and sharing
- **Group Policy Management:** Centralized configuration management

6.2 Domain Configuration

- **Domain Name:** mycompany.local
- **Forest Functional Level:** Windows Server 2022
- **Domain Controllers:** SRV-2022 (primary)

7. Security Enhancements Implemented

7.1 Access Control

- **Group-based permissions:** Simplifies management and reduces errors
- **Least privilege principle:** Users only access necessary resources
- **Regular permission auditing:** Enabled through NTFS auditing features

7.2 Physical Security

- **USB restriction policy:** Prevents data exfiltration via removable media
- **Automatic workstation locking:** Prevents unauthorized access to unattended computers

7.3 Password Security

- **Enforced complexity and rotation:** Through password policy GPO
- **Account lockout policies:** Protects against brute force attacks

8. Alignment with Business Requirements

8.1 Data Security

The implementation addresses the accidental customer file leakage incident through:

- Centralized file storage with access controls
- USB device restrictions
- Audit logging capabilities

8.2 Document Centralization

The shared folder structure provides:

- Departmental storage spaces with appropriate access rights
- Common area for cross-department collaboration
- Personal user directories (home folders)

8.3 Simplified User Management

The implementation enables HR to:

- Use PowerShell scripts for standardized account creation
- Manage users within clearly organized OUs
- Assign permissions through group membership

8.4 Digital Sobriety (DDRS)

The infrastructure supports sustainability through:

- Server virtualization capability
- Centralized storage reducing local disk usage
- Potential for managed printing policies via GPO

10. Testing and Validation

10.1 Access Rights Validation

Each folder's permissions were tested by:

- Logging in as users from different departments
- Verifying read access where specified
- Verifying write access where specified
- Confirming access denial where appropriate

10.2 GPO Application Testing

Password policy enforcement was validated by:

- Creating test accounts with weak passwords (blocked)
- Verifying password expiration notices appear
- Testing password history enforcement

10.3 Script Testing

PowerShell scripts were tested by:

- Creating multiple test user accounts
- Verifying correct OU placement
- Checking group membership assignment
- Confirming permission application

11. Documentation and Training

11.1 Administrator Documentation

Complete documentation was prepared including:

- AD structure diagrams
- Permission matrices
- PowerShell script usage guides
- GPO configuration details

- Troubleshooting procedures

11.2 End-User Training

Supporting Deliverable 2 (CAG):

- How to access shared folders
- Password requirements and best practices
- USB policy restrictions
- Automatic lock screen usage
- Reporting security incidents

12. Future Recommendations

12.1 Short-term Improvements

- Enable password complexity requirements in GPO
- Implement account lockout policy
- Deploy additional security GPOs (USB restriction, auto-lock)
- Configure automated backup solution for AD and file shares

12.2 Long-term Enhancements

- Implement Multi-Factor Authentication (MFA)
- Deploy Data Loss Prevention (DLP) solution
- Consider Azure AD integration for cloud services
- Implement Security Information and Event Management (SIEM)

The infrastructure delivers a secure foundation for GreenReso through Active Directory with departmental OUs, shared folders with NTFS permissions, and centralized document management. PowerShell automation streamlines user account management, while GPOs enforce consistent security policies, addressing gaps and supporting growth. The scalable design accommodates expansion and upcoming migrations without major architectural changes.